

Mehul Kaushik

8929846800 | mehul.kaushik6.cse@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [LeetCode](#)

EDUCATION

Manipal Institute of Technology
Bachelor of Technology in Computer Science

Manipal, India
2023 – 2027

EXPERIENCE

Project Intern
STMicroelectronics

May 2025 – July 2025
Noida, India

- Automated reporting workflows with Python, reducing manual effort by 60%.
- Built modular scripts for the CVIP PLL verification layer, cutting verification work by 40% and doubling regression speed.

PROJECTS

MoonMovement – Startup Community Platform | *PostgreSQL, Prisma, Node.js, Redis, AWS* [GitHub](#) | [Live](#)

- Built a Reddit-style community platform with startup discovery, threaded discussions, voting, user profiles, directories, and role-based moderation.
- Designed and deployed a three-tier backend on AWS EC2 with stateless APIs, PostgreSQL on RDS, Redis caching, and an Nginx reverse proxy.
- Implemented session management, feed caching, and rate limiting; optimized Prisma queries to eliminate N+1 lookups and lower database load.
- Implemented authentication, RBAC, asynchronous notifications, database indexing, pagination, and transactional consistency.

RAG-Powered Document Q&A System | *Node.js, MongoDB, Redis, Vector Search*

[GitHub](#)

- Enabled users to upload and query PDFs through a conversational LLM interface.
- Created a five-stage asynchronous pipeline (parse → chunk → embed → index → search) for documents up to 20 MB, generating 768-dimensional embeddings for semantic retrieval.
- Offloaded processing to a Redis worker using a BRPOP queue, allowing the API to return 202 Accepted while tracking job status with hashes (HSET).
- Reduced hallucinations with a similarity threshold (≥ 0.7) and added state cleanup to prevent orphaned MongoDB records after processing failures.

TECHNICAL SKILLS

Languages: C++, Python, Java, SQL

Backend: Node.js, Express.js, REST APIs, PostgreSQL, MongoDB, Prisma, Redis

Cloud & DevOps: AWS (EC2, RDS), Docker, Nginx, Git, Linux

Machine Learning & AI: RAG, LLM applications, NLP, vector search, CNN

ACHIEVEMENTS

- Solved 600+ LeetCode problems and earned the Guardian badge.
- Ranked 318th nationally in the Amazon Machine Learning Challenge.
- Won three hackathons and was selected as a Smart India Hackathon (SIH) finalist.
- Contributed one merged pull request to the HumanAI project during Google Summer of Code.

CERTIFICATIONS

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| Machine Learning Specialization by Andrew Ng | Credential |
| Google AI Professional Certificate | Credential |
| Google Data Analytics Professional Certificate | Credential |
| AWS Cloud Practitioner Essentials | Credential |